

OUTCOMES

The Multi-Source AI for Hyperlocal Rainfall Nowcasting system successfully established the core software and data-processing framework for localized rainfall intelligence. The project demonstrates the integration of multi-source weather observations, automated data acquisition, data processing and fusion, AI/ML-based short-term rainfall nowcasting, numerical weather prediction processing and a web-based visualization platform.

- Successfully integrated weather observations from IMD Automatic Weather Stations (AWS), Personal Weather Stations (PWS), and other supporting weather-data sources.
- Implemented automated weather-data acquisition using API-based data fetchers and scheduled collection mechanisms for continuously updated observations.
- Established a structured weather-data storage layer using SQLite during local development, with support for PostgreSQL-based production deployment.
- Implemented data cleaning, validation, timestamp alignment and multi-source data fusion to prepare weather observations for analysis and prediction.
- Generated important meteorological features such as pressure trend and drop rate, humidity trend, dew-point spread, previous rainfall and spatial neighbouring-station information.
- Developed the AI/ML rainfall nowcasting pipeline using suitable machine-learning and deep-learning approaches, with the objective of generating short-term rainfall predictions for the next 0–3 hours.
- Integrated ECMWF Numerical Weather Prediction (NWP) data processing for longer-range forecast information and approximately 1–10 day forecast visualization.
- Implemented a FastAPI-based backend and REST API layer to connect weather-data sources, storage, processing, prediction services and the web application.
- Developed a web-based interface using Next.js, React and TypeScript for displaying real-time weather observations, rainfall nowcasts, graphs, station comparisons and forecast information.
- Established an end-to-end workflow connecting weather-data acquisition, storage, processing, AI/ML prediction, NWP processing, API services and user-facing visualization.
- Provided a unified platform for localized weather monitoring, short-term rainfall nowcasting and longer-range forecast visualization for the Puducherry and Tamil Nadu region.
- Created a foundation for future enhancements including real-time weather radar integration, advanced spatiotemporal deep-learning models, improved prediction performance, severe-weather alerts and wider geographic coverage.

Overall, the completed project establishes the core architecture and implementation workflow for multi-source hyperlocal rainfall intelligence, combining real-time observations, AI-based 0–3 hour nowcasting and ECMWF-based longer-range forecast visualization in a single platform.